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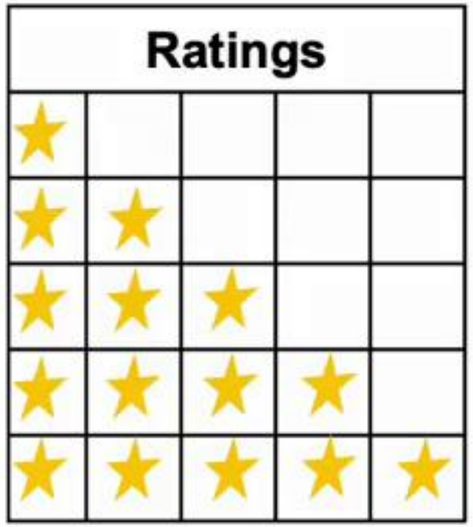
Recommender System

Making recommendations

Predicting movie ratings

User rates movies using ~~one~~ zero to five stars

Movie	Alice(1)	Bob(2)	Carol(3)	Dave(4)
Love at last	5	5	0	0
Romance forever	5	?	?	0
Cute puppies of love	?	4	0	?
Nonstop car chases	0	0	5	4
Swords vs. karate	0	0	5	?



n_u = no. of users

n_m = no. of movies

$r(i,j)=1$ if user j has rated movie i

$y^{(i,j)}$ = rating given by user j to movie i (defined only if $r(i,j)=1$)

$n_u = 4$ $\quad r(1,1) = 1$

$n_m = 5$ $\quad r(3,1) = 0$

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Collaborative Filtering

Using per-item features

What if we have features of the movies?

$n_u = 4$
$n_m = 5$
$n = 2$

Movie	Alice(1)	Bob(2)	Carol(3)	Dave(4)	x_1 (romance)	x_2 (action)
Love at last	5	5	0	0	0.9	0
Romance forever	5	?	?	0	1.0	0.01
Cute puppies of love	?	4	0	?	0.99	0
Nonstop car chases	0	0	5	4	0.1	1.0
Swords vs. karate	0	0	5	?	0	0.9

$$x^{(1)} = \begin{bmatrix} 0.9 \\ 0 \end{bmatrix}$$

$$x^{(3)} = \begin{bmatrix} 0.99 \\ 0 \end{bmatrix}$$

For user 1: Predict rating for movie *i* as: $w^{(1)} \cdot x^{(i)} + b^{(1)}$

just linear regression

$w^{(1)} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$ $b^{(1)} = 0$ $x^{(3)} = \begin{bmatrix} 0.99 \\ 0 \end{bmatrix}$

$w^{(1)} \cdot x^{(3)} + b^{(1)} = \underline{4.95}$

For user *j*: Predict user j's rating for movie *i* as $w^{(j)} \cdot x^{(i)} + b^{(j)}$

Cost function

Notation:

$r(i,j) = 1$ if user j has rated movie i (0 otherwise)
$y^{(i,j)}$ = rating given by user j on movie i (if defined)
$w^{(j)}, b^{(j)}$ = parameters for user j
$x^{(i)}$ = feature vector for movie i

For user j and movie i, predict rating: $w^{(j)} \cdot x^{(i)} + b^{(j)}$
$m^{(j)}$ = no. of movies rated by user j
To learn $w^{(j)}, b^{(j)}$

number of features

$$\min_{w^{(j)}, b^{(j)}} J(w^{(j)}, b^{(j)}) = \frac{1}{2m^{(j)}} \sum_{i:r(i,j)=1} \left(w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2m^{(j)}} \sum_{k=1}^{n} \left(w_k^{(j)}\right)^2$$

Cost function

To learn parameters $w^{(j)}, b^{(j)}$ for user j :

$$J\left(w^{(j)}, b^{(j)}\right) = \frac{1}{2} \sum_{i:r(i,j)=1} \left(w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2} \sum_{k=1}^{n} \left(w_k^{(j)}\right)^2$$

To learn parameters $w^{(1)}, b^{(1)},\ w^{(2)}, b^{(2)}, \cdots\ w^{(n_u)}, b^{(n_u)}$ for all users :

$$J\begin{pmatrix} w^{(1)}, & \ldots, w^{(n_u)} \\ b^{(1)}, & \ldots, b^{(n_u)} \end{pmatrix} = \frac{1}{2} \sum_{j=1}^{n_u} \sum_{i:r(i,j)=1} \left(w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2} \sum_{j=1}^{n_u} \sum_{k=1}^{n} \left(w_k^{(j)}\right)^2$$

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Collaborative Filtering

Collaborative filtering algorithm

Problem motivation

Movie	Alice (1)	Bob (2)	Carol (3)	Dave (4)	x_1 (romance)	x_2 (action)
Love at last	5	5	0	0	?	?
Romance forever	5	?	?	0	?	? $x^{(2)}$
Cute puppies of love	?	4	0	?	?	?
Nonstop car chases	0	0	5	4	?	?
Swords vs. karate	0	0	5	?	?	?

$$w^{(1)}=\begin{bmatrix}5\\0\end{bmatrix},\ w^{(2)}=\begin{bmatrix}5\\0\end{bmatrix},\ w^{(3)}=\begin{bmatrix}0\\5\end{bmatrix},\ w^{(4)}=\begin{bmatrix}0\\5\end{bmatrix}$$

$$b^{(1)}=0,\ b^{(2)}=0,\ b^{(3)}=0,\ b^{(4)}=0$$

using $\boldsymbol{w}^{(j)}\cdot\boldsymbol{x}^{(i)}+\cancel{\boldsymbol{b}^{(j)}}$

$$\left.\begin{aligned} w^{(1)}\cdot x^{(1)} &\approx 5\\ w^{(2)}\cdot x^{(1)} &\approx 5\\ w^{(3)}\cdot x^{(1)} &\approx 0\\ w^{(4)}\cdot x^{(1)} &\approx 0 \end{aligned}\right\} \rightarrow \quad x^{(1)}=\begin{bmatrix}1\\0\end{bmatrix}$$

Cost function

Given $w^{(1)}, b^{(1)}, w^{(2)}, b^{(2)}, \cdots, w^{(n_u)}, b^{(n_u)}$

to learn $x^{(i)}$:

$$J(x^{(i)}) = \frac{1}{2} \sum_{j:r(i,j)=1} \left(w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2} \sum_{k=1}^{n} \left(x_k^{(i)}\right)^2$$

→ To learn $x^{(1)}, x^{(2)}, \cdots, x^{(n_m)}$:

$$J(x^{(1)}, x^{(2)}, \ldots, x^{(n_m)}) = \frac{1}{2} \sum_{i=1}^{n_m} \sum_{j:r(i,j)=1} \left(w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2} \sum_{i=1}^{n_m} \sum_{k=1}^{n} \left(x_k^{(i)}\right)^2$$

Collaborative filtering

	$j=1$ Alice	$j=2$ Bob	$j=3$ Carol
$i=1$ Movie1	5	5	?
$i=2$ Movie2	?	2	3

Cost function to learn $w^{(1)}, b^{(1)}, \cdots w^{(n_u)}, b^{(n_u)}$:

$$\min_{w^{(1)},b^{(1)},\ \cdots,\ w^{(n_u)},b^{(n_u)}} \frac{1}{2}\sum_{j=1}^{n_u}\sum_{i:r(i,j)=1}\left(w^{(j)}\cdot x^{(i)}+b^{(j)}-y^{(i,j)}\right)^2 + \frac{\lambda}{2}\sum_{j=1}^{n_u}\sum_{k=1}^{n}\left(w_k^{(j)}\right)^2$$

Cost function to learn $x^{(1)}, \cdots, x^{(n_m)}$:

$$\min_{x^{(1)},\ \cdots,\ x^{(n_m)}} \frac{1}{2}\sum_{i=1}^{n_m}\sum_{j:r(i,j)=1}\left(w^{(j)}\cdot x^{(i)}+b^{(j)}-y^{(i,j)}\right)^2 + \frac{\lambda}{2}\sum_{i=1}^{n_m}\sum_{k=1}^{n}\left(x_k^{(i)}\right)^2$$

Put them together:

$$\min_{\substack{w^{(1)},\ \ldots,\ w^{(n_u)}\\ b^{(1)},\ \ldots,\ b^{(n_u)}\\ x^{(1)},\ \ldots,\ x^{(n_m)}}} J(w,b,x) = \frac{1}{2}\sum_{(i,j):r(i,j)=1}\left(w^{(j)}\cdot x^{(i)}+b^{(j)}-y^{(i,j)}\right)^2 + \frac{\lambda}{2}\sum_{j=1}^{n_u}\sum_{k=1}^{n}\left(w_k^{(j)}\right)^2 + \frac{\lambda}{2}\sum_{i=1}^{n_m}\sum_{k=1}^{n}\left(x_k^{(i)}\right)^2$$

Gradient Descent

collaborative filtering

Linear regression (course 1)

repeat {

$$\cancel{w_i = w_i - \alpha \frac{\partial}{\partial w_i} J(w,b)}$$

$$\cancel{b = b - \alpha \frac{\partial}{\partial b} J(w,b)}$$

$$w_i^{(j)} = w_i^{(j)} - \alpha \frac{\partial}{\partial w_i^{(j)}} J(w,b,x)$$

$$b^{(j)} = b^{(j)} - \alpha \frac{\partial}{\partial b^{(j)}} J(w,b,x)$$

$$x_k^{(i)} = x_k^{(i)} - \alpha \frac{\partial}{\partial x_k^{(i)}} J(w,b,x)$$

}

parameters w, b, x

x is also a parameter

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Collaborative Filtering

Binary labels:
favs,
likes and clicks

Binary labels

Movie	Alice(1)	Bob(2)	Carol(3)	Dave(4)
Love at last	1	1	0	0
Romance forever	1	?	?	0
Cute puppies of love	?	1	0	?
Nonstop car chases	0	0	1	1
Swords vs. karate	0	0	1	?

Example applications

1. Did user j purchase an item after being shown?
2. Did user j fav/like an item?
3. Did user j spend at least 30sec with an item?
4. Did user j click on an item?

Meaning of ratings:
1 - engaged after being shown item
0 - did not engage after being shown item
? - item not yet shown

From regression to binary classification

Previously:

Predict $\boldsymbol{y}^{(i,j)}$ as $\boldsymbol{w}^{(j)} \cdot \boldsymbol{x}^{(i)} + \boldsymbol{b}^{(j)}$

For binary labels:

Predict that the probability of $\boldsymbol{y}^{(i,j)} = \mathbf{1}$ is given by $g\left(\boldsymbol{w}^{(j)} \cdot \boldsymbol{x}^{(i)} + \boldsymbol{b}^{(j)}\right)$

where $g(z) = \frac{1}{1+e^{-z}}$

Cost function for binary application

Previous cost function:

$$\frac{1}{2}\sum_{(i,j):r(i,j)=1}\left(\underbrace{w^{(j)}\cdot x^{(i)}+b^{(j)}}_{f(x)}-y^{(i,j)}\right)^2+\frac{\lambda}{2}\sum_{i=1}^{n_m}\sum_{k=1}^{n}\left(x_k^{(i)}\right)^2+\frac{\lambda}{2}\sum_{j=1}^{n_u}\sum_{k=1}^{n}\left(w_k^{(j)}\right)^2$$

Loss for binary labels $y^{(i,j)}$: $\boxed{f_{(w,b,x)}(x)=g(w^{(j)}\cdot x^{(i)}+b^{(j)})}$

$$L\left(f_{(w,b,x)}(x),y^{(i,j)}\right)=-y^{(i,j)}\log\left(f_{(w,b,x)}(x)\right)-\left(1-y^{(i,j)}\right)\log\left(1-f_{(w,b,x)}(x)\right)$$

Loss for single example

$$J(w,b,x)=\sum_{(i,j):r(i,j)=1}L\left(\underbrace{f_{(w,b,x)}(x)}_{g(w^{(j)}\cdot x^{(i)}+b^{(j)})},y^{(i,j)}\right)$$

cost for all examples

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Recommender Systems implementation

Mean normalization

Users who have not rated any movies

Movie	Alice(1)	Bob (2)	Carol (3)	Dave (4)	Eve (5)	
Love at last	5	5	0	0	?	0
Romance forever	5	?	?	0	?	0
Cute puppies of love	?	4	0	?	?	0
Nonstop car chases	0	0	5	4	?	0
Swords vs. karate	0	0	5	?	?	0

$$\begin{bmatrix} 5 & 5 & 0 & 0 & ? \\ 5 & ? & ? & 0 & ? \\ ? & 4 & 0 & ? & ? \\ 0 & 0 & 5 & 4 & ? \\ 0 & 0 & 5 & 0 & ? \end{bmatrix}$$

$$\min_{\substack{w^{(1)},\ldots w^{(n_u)} \\ b^{(1)},\ldots b^{(n_u)} \\ x^{(1)},\ldots x^{(n_m)}}} \frac{1}{2} \sum_{(i,j):r(i,j)=1} \left(w^{(j)} \cdot x^{(i)} + b^{(j)} - y^{(i,j)}\right)^2 + \frac{\lambda}{2} \sum_{j=1}^{n_u} \sum_{k=1}^{n} \left(w_k^{(j)}\right)^2 + \frac{\lambda}{2} \sum_{i=1}^{n_m} \sum_{k=1}^{n} \left(x_k^{(i)}\right)^2$$

$$w^{(5)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad b^{(5)} = 0 \qquad w^{(5)} \cdot x^{(i)} + b^{(5)}$$

Mean Normalization

$$\begin{bmatrix} 5 & 5 & 0 & 0 & ? \\ 5 & ? & ? & 0 & ? \\ ? & 4 & 0 & ? & ? \\ 0 & 0 & 5 & 4 & ? \\ 0 & 0 & 5 & 0 & ? \end{bmatrix} \begin{matrix} 2.5 \\ 2.5 \\ 2 \\ 2.25 \\ 1.25 \end{matrix} \qquad \mu = \begin{bmatrix} 2.5 \\ 2.5 \\ 2 \\ 2.25 \\ 1.25 \end{bmatrix} \qquad \underbrace{\begin{bmatrix} 2.5 & 2.5 & -2.5 & -2.5 & ? \\ 2.5 & ? & ? & -2.5 & ? \\ ? & 2 & -2 & ? & ? \\ -2.25 & -2.25 & 2.75 & 1.75 & ? \\ -1.25 & -1.25 & 3.75 & -1.25 & ? \end{bmatrix}}_{y^{(i,j)}}$$

$$w^{(j)} \cdot b^{(j)} \cdot x^{(i)}$$

For user j, on movie i predict:

$$w^{(j)} \cdot x^{(i)} + b^{(j)} + \mu_i$$

User 5 (Eve):

$$w^{(5)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad b^{(5)} = 0 \qquad \underbrace{w^{(5)} \cdot x^{(1)} + b^{(5)}}_{0} + \mu_1$$

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Recommender Systems implementational detail

TensorFlow implementation

Derivatives in ML

Gradient descent algorithm

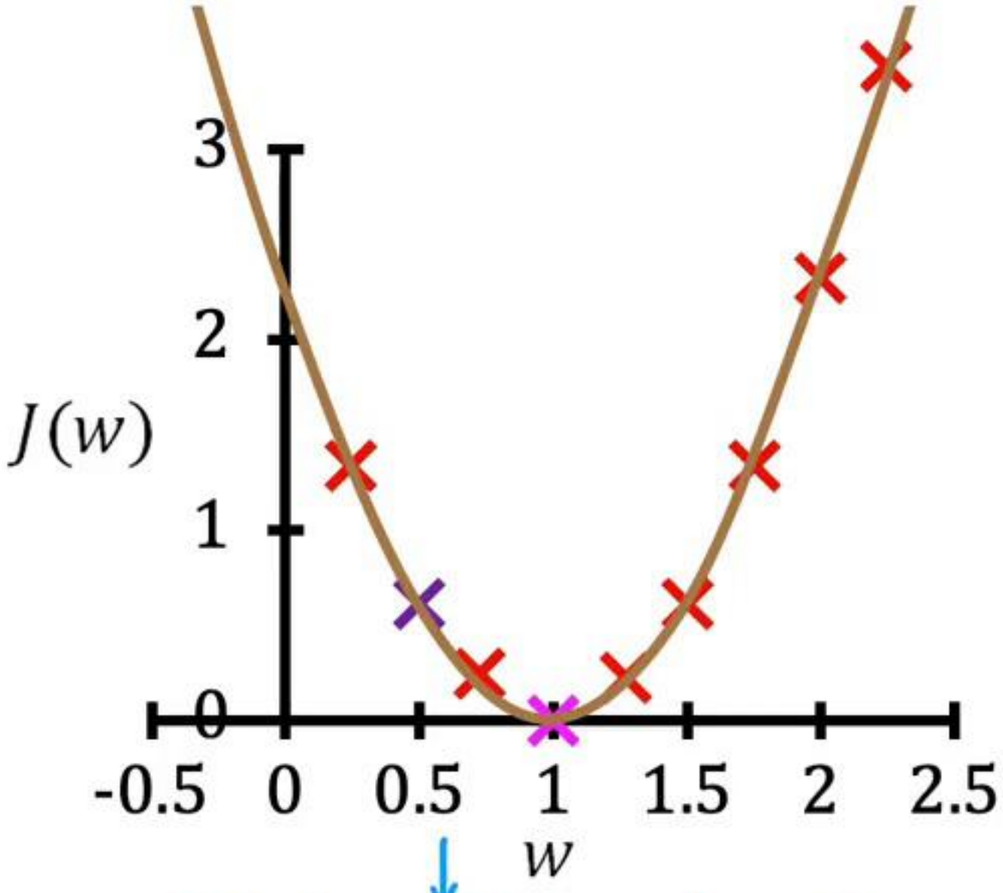
Repeat until convergence

$$w = w - \alpha \frac{\partial}{\partial w} J(w,b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w,b) \leftarrow b = 0$$

Learning rate

Derivative



$f(x) = w \cdot x \quad b = 0$

Custom Training Loop

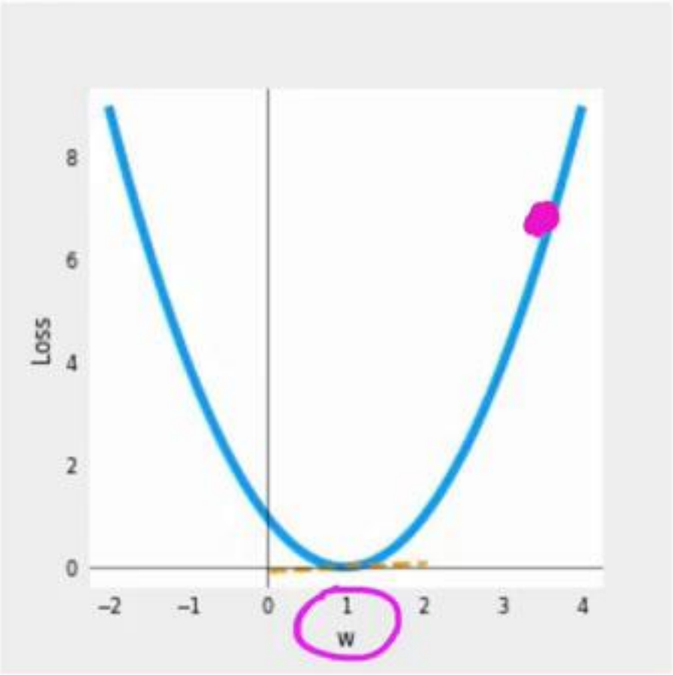
$$J = (\overbrace{wx}^{f(x)} - \overbrace{1}^{y})^2$$

Gradient descent algorithm

Repeat until convergence

$$w = w - \alpha \frac{\partial}{\partial w} J(w,b)$$

Fix b = 0 for this example




```
w = tf.Variable(3.0)
x = 1.0
y = 1.0 # target value
alpha = 0.01

iterations = 30
for iter in range(iterations):
    # Use TensorFlow's Gradient tape to record the steps
    # used to compute the cost J, to enable auto differentiation.
    with tf.GradientTape() as tape:
        fwb = w*x
        costJ = (fwb - y)**2

    # Use the gradient tape to calculate the gradients
    # of the cost with respect to the parameter w.
    [dJdw] = tape.gradient( costJ, [w] )

    # Run one step of gradient descent by updating
    # the value of w to reduce the cost.
    w.assign_add(-alpha * dJdw)
```

f(x)

Tf.variables are the parameters we want to optimize

Auto Diff

Auto Grad

$$\frac{\partial}{\partial w} J(w)$$

tf.variables require special function to modify

Implementation in TensorFlow

Gradient descent algorithm

Repeat until convergence

$$w = w - \alpha \frac{\partial}{\partial w} J(w,b,X)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w,b,X)$$

$$X = X - \alpha \frac{\partial}{\partial X} J(w,b,X)$$

```
# Instantiate an optimizer.
optimizer = keras.optimizers.Adam(learning_rate=1e-1)

iterations = 200
for iter in range(iterations):
    # Use TensorFlow's GradientTape
    # to record the operations used to compute the cost
    with tf.GradientTape() as tape:

        # Compute the cost (forward pass is included in cost)
        cost_value = cofiCostFuncV(X, W, b, Ynorm, R,
            num_users, num_movies, lambda)
```

n_u n_m

```
    # Use the gradient tape to automatically retrieve
    # the gradients of the trainable variables with respect to
        the loss
    grads = tape.gradient( cost_value, [X,W,b] )

    # Run one step of gradient descent by updating
    # the value of the variables to minimize the loss.
    optimizer.apply_gradients( zip(grads, [X,W,b]) )
```

Dataset credit: Harper and Konstan. 2015. The MovieLens Datasets: History and Context

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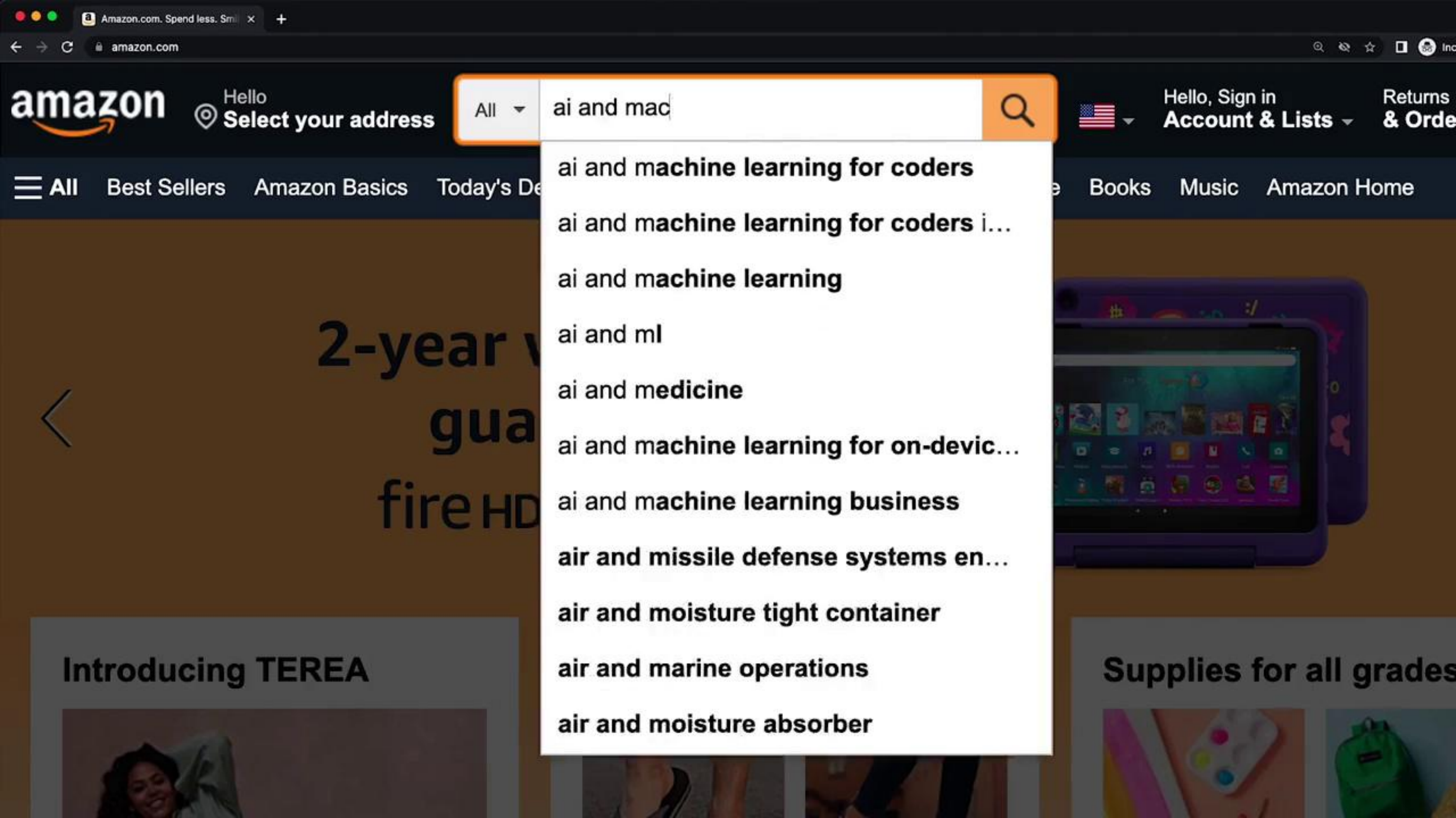
Collaborative Filtering

Finding related items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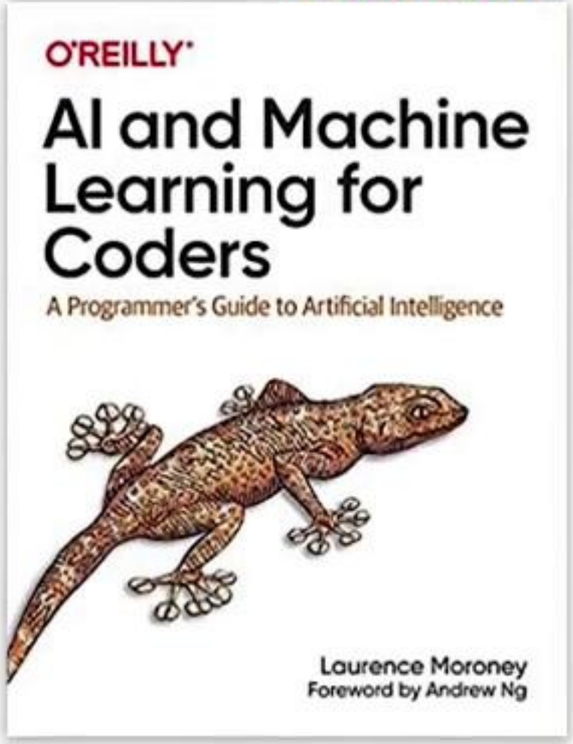

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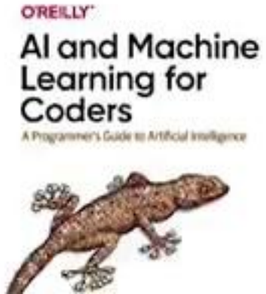

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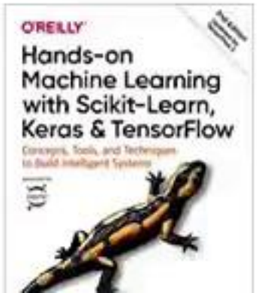

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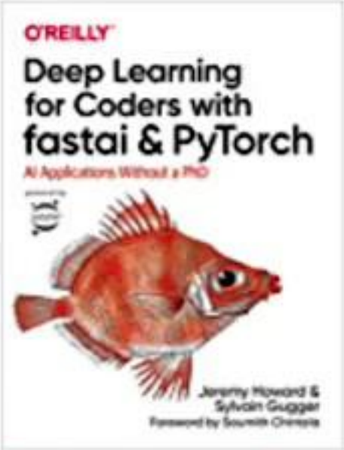

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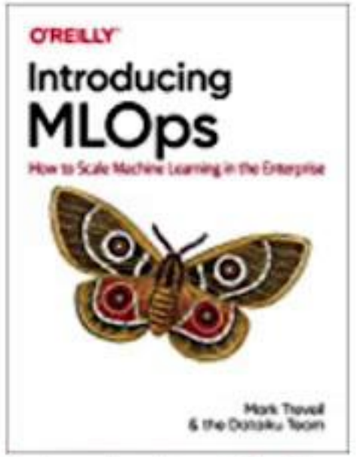

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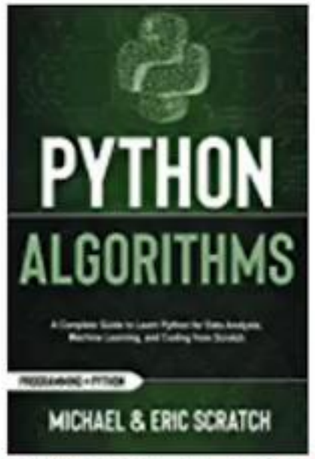

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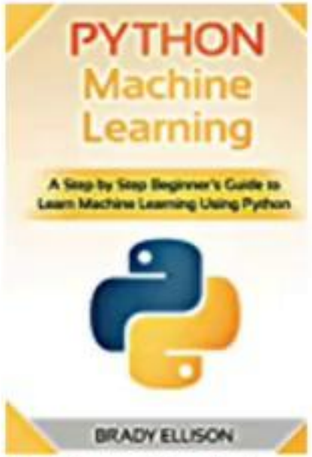

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SUGGESTED EXTRAS DETAILS

Finding related items

romance

action

The features $x^{(i)}$ of item i are quite hard to interpret.

To find other items related to it,

find item k with $x^{(k)}$ similar to $x^{(i)}$

x_1, x_2, x_3

n

i.e. with smallest distance

$$\sum_{l=1}^{n} \left(x_l^{(k)} - x_l^{(i)} \right)^2$$

$$\left\| \boldsymbol{x}^{(k)} - \boldsymbol{x}^{(i)} \right\|^2$$

Limitations of Collaborative Filtering

Cold start problem. How to

- rank new items that few users have rated?
- show something reasonable to new users who have rated few items?

Use side information about items or users:

- Item: Genre, movie stars, studio, ….
- User: Demographics (age, gender, location), expressed preferences, …

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Content-based Filtering

Collaborative filtering
vs
Content-based filtering

Collaborative filtering vs Content-based filtering

- Collaborative filtering:
  Recommend items to you based on ratings of users who gave similar ratings as you

- Content-based filtering:
  Recommend items to you based on features of user and item to find good match

$r(i,j) = 1$ if user j has rated item i

$y^{(i,j)}$ rating given by user j on item i (if defined)

Examples of user and item features

User features:

- Age
- Gender (1 hot)
- Country (1 hot, 200)
- Movies watched (1000)
- Average rating per genre
- …

$\mathbf{x}_u^{(j)}$ **for user j**

Vector size could be different

Movie features:

- Year
- Genre/Genres
- Reviews
- Average rating
- …

$\mathbf{x}_m^{(i)}$ **for movie i**

Content-based filtering: Learning to match

Predict rating of user *j* on movie *i* as

$$w^{(j)} \cdot x^{(i)} + \cancel{b^{(j)}}$$

$$v_u^{(j)} \cdot v_m^{(i)}$$

computed from $\boldsymbol{x}_u^{(j)}$

computed from $\boldsymbol{x}_m^{(i)}$

$x_u^{(j)}$

user's preferences

$$v_u^{(j)} = \begin{bmatrix} 4.9 \\ 0.1 \\ \vdots \\ 3.0 \end{bmatrix} \begin{matrix} \text{likes} \\ \text{likes} \\ \\ \\ \end{matrix}$$

32

$x_m^{(i)}$

movie features

$$v_m^{(i)} = \begin{bmatrix} 4.5 \\ 0.2 \\ \vdots \\ 3.5 \end{bmatrix} \begin{matrix} \text{romance} \\ \text{action} \\ \\ \\ \end{matrix}$$

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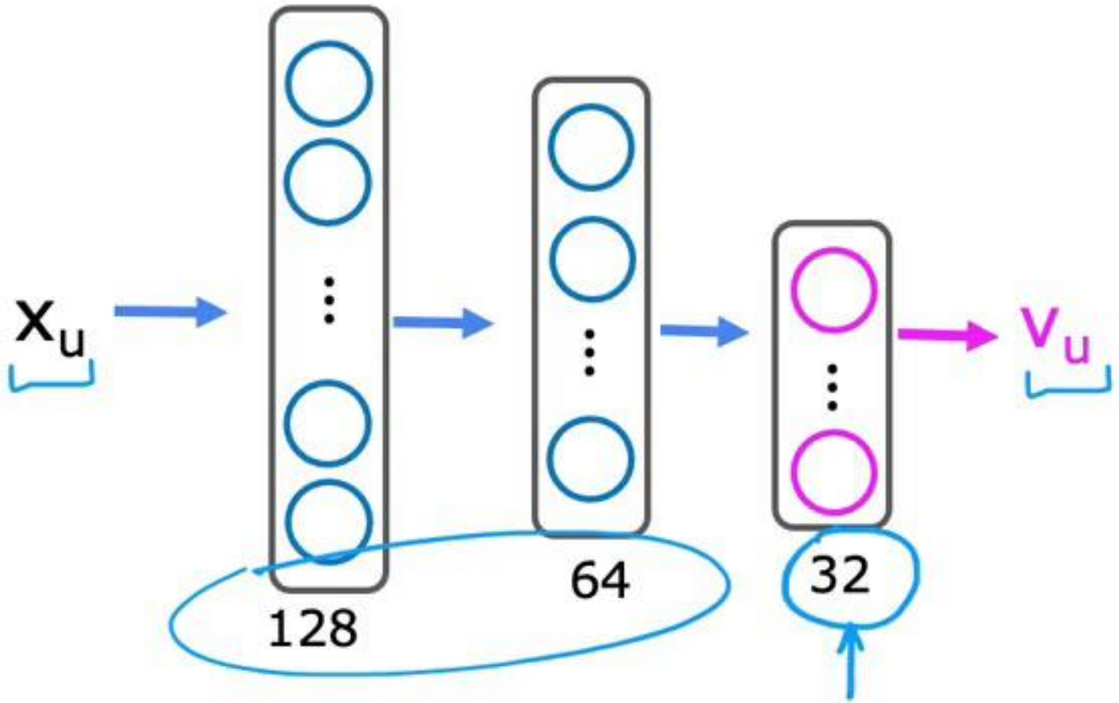
Content-based Filtering

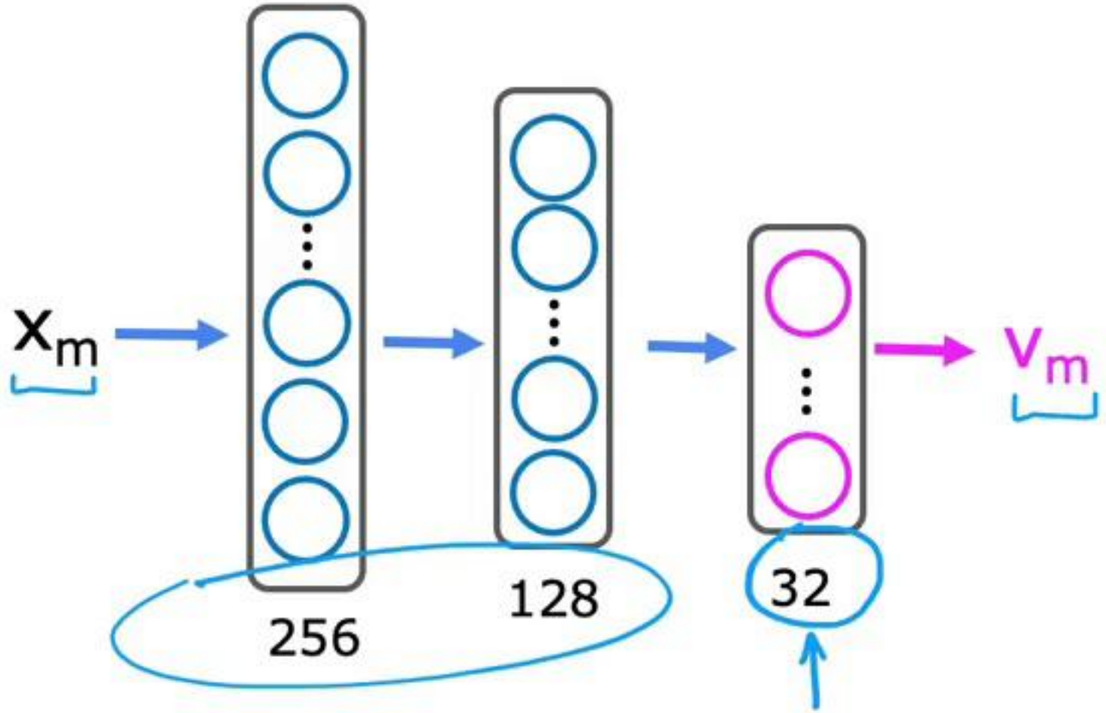
Deep learning for content-based filtering

Neural network architecture

$x_u \rightarrow v_u$ User network



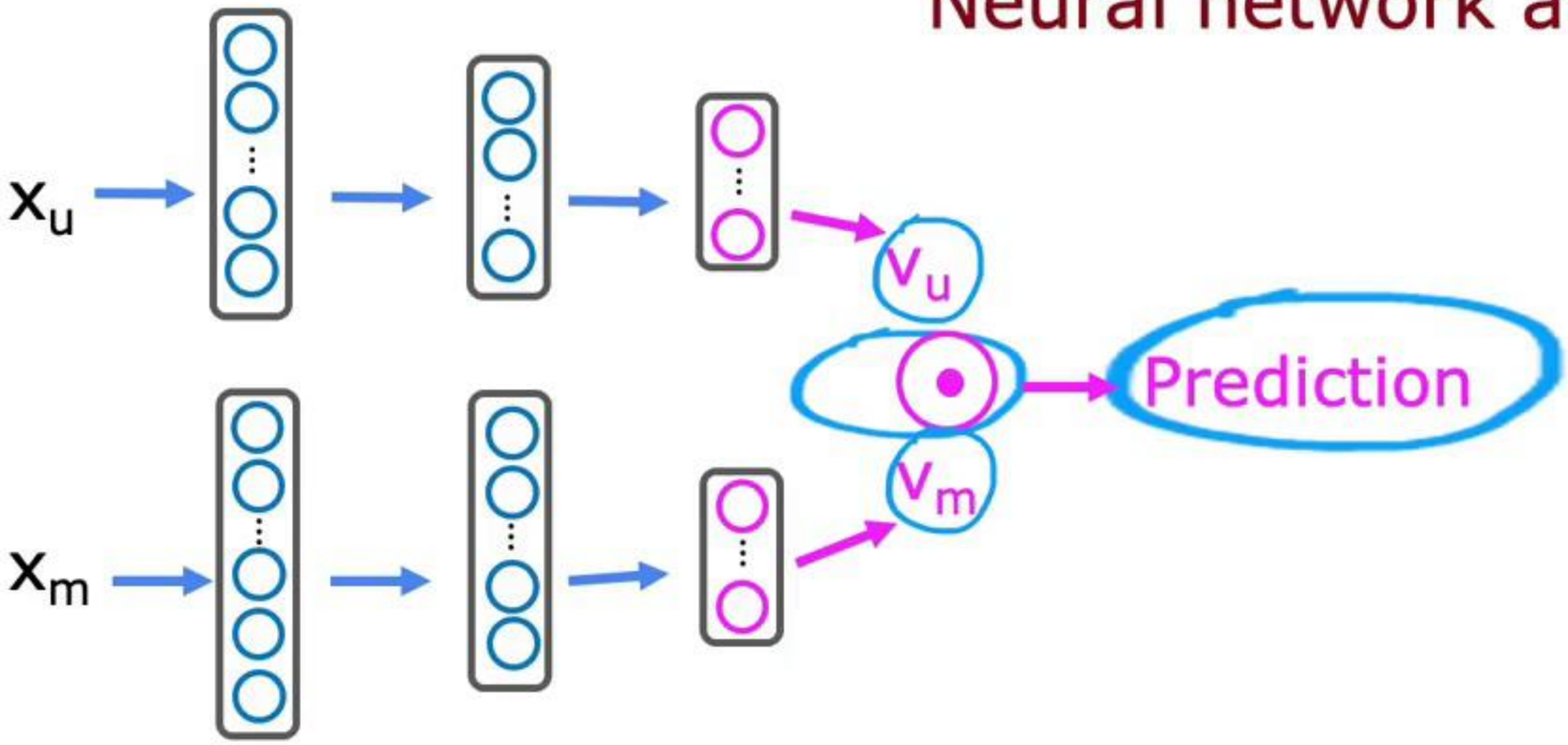
$x_m \rightarrow v_m$ Movie network



Prediction : $v_u \cdot v_m$

$g(v_u \cdot v_m)$ *to predict the probability that* $y^{(i,j)}$ *is* 1

Neural network architecture



Cost function $$J = \sum_{(i,j):r(i,j)=1} \left(v_u^{(j)} \cdot v_m^{(i)} - y^{(i,j)}\right)^2 + \text{NN regularization term}$$

Learned user and item vectors:

$\boldsymbol{v}_u^{(j)}$ is a vector of length 32 that describes user j with features $\boldsymbol{x}_u^{(j)}$

$\boldsymbol{v}_m^{(i)}$ is a vector of length 32 that describes movie i with features $\boldsymbol{x}_m^{(i)}$

To find movies similar to movie i: $\|v_m^{(k)} - v_m^{(i)}\|^2$ small

$\|x^{(k)} - x^{(i)}\|^2$

Note: This can be pre-computed ahead of time

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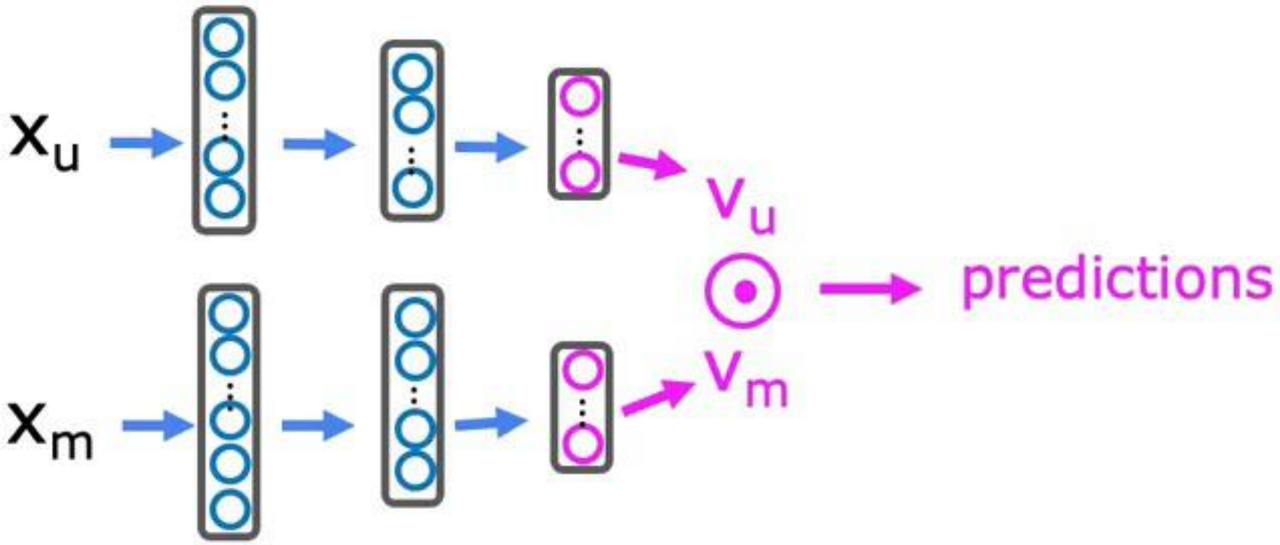
Advanced implementation

Recommending from a large catalogue

How to efficiently find recommendation from a large set of items?

- Movies 1000+
- Ads 1m+
- Songs 10m+
- Products 10m+



Two steps: Retrieval & Ranking

Retrieval:

- Generate large list of plausible item candidates ~100s

  e.g.

  1) For each of the last 10 movies watched by the user, find 10 most similar movies

     $\| v_m^{(k)} - v_m^{(i)} \|^2$

  2) For most viewed 3 genres, find the top 10 movies
  3) Top 20 movies in the country

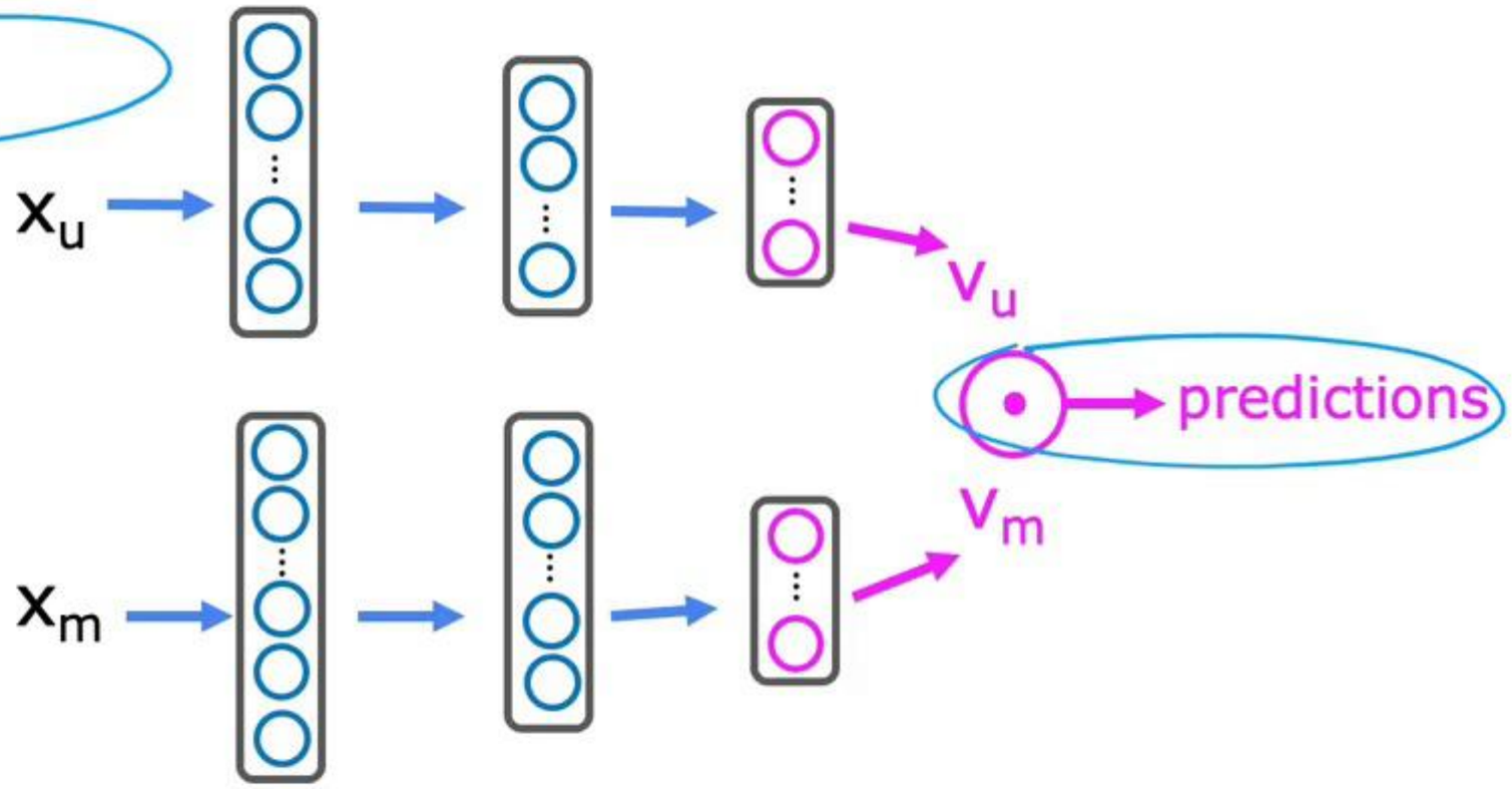
- Combine retrieved items into list, removing duplicates and items already watched/purchased

Two steps: Retrieval & ranking

Ranking:

~100s

- Take list retrieved and rank using learned model



- Display ranked items to user

Retrieval step

- Retrieving more items results in better performance, but slower recommendations.
- To analyse/optimize the trade-off, carry out offline experiments to see if retrieving additional items results in more relevant recommendations (i.e., $p\left(y^{(i,j)}\right) = 1$ of items displayed to user are higher).

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Advanced implementation

Ethical use of recommender systems

What is the goal of the recommender system?

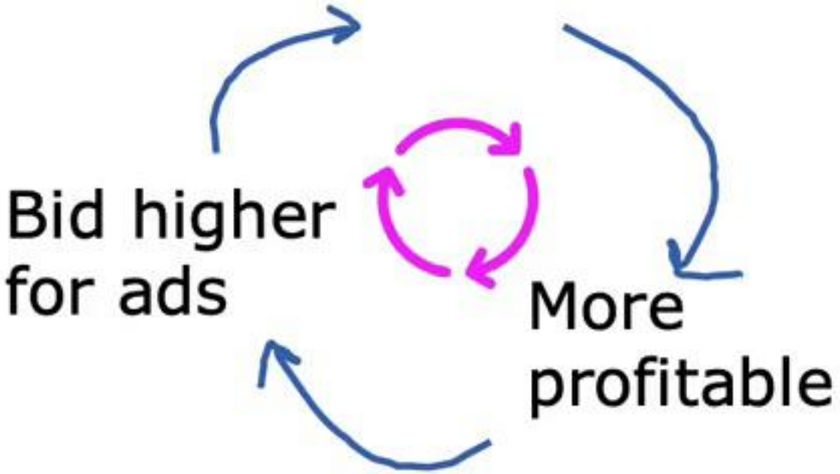
Recommend:

- Movies most likely to be rated 5 stars by user
- Products most likely to be purchased
- Ads most likely to be clicked on + high bid
- Products generating the largest profit
- Video leading to maximum watch time

Ethical considerations with recommender systems

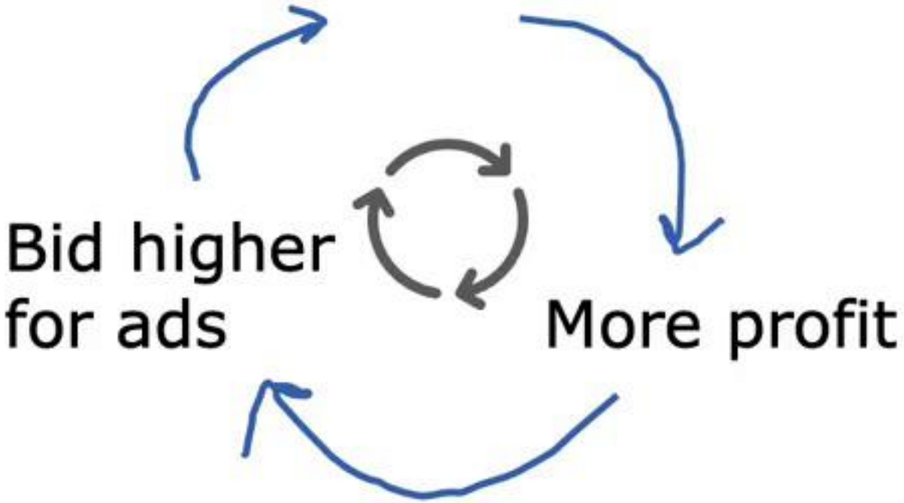
Travel industry

Good travel experience to more users



Payday loans

Squeeze customers more



Amelioration: Do not accept ads from exploitative businesses

Other problematic cases:

- Maximizing user engagement (e.g. watch time) has led to large social media/video sharing sites to amplify conspiracy theories and hate/toxicity

  Amelioration : Filter out problematic content such as hate speech, fraud, scams and violent content

- Can a ranking system maximize your profit rather than users' welfare be presented in a transparent way?

  Amelioration : Be transparent with users

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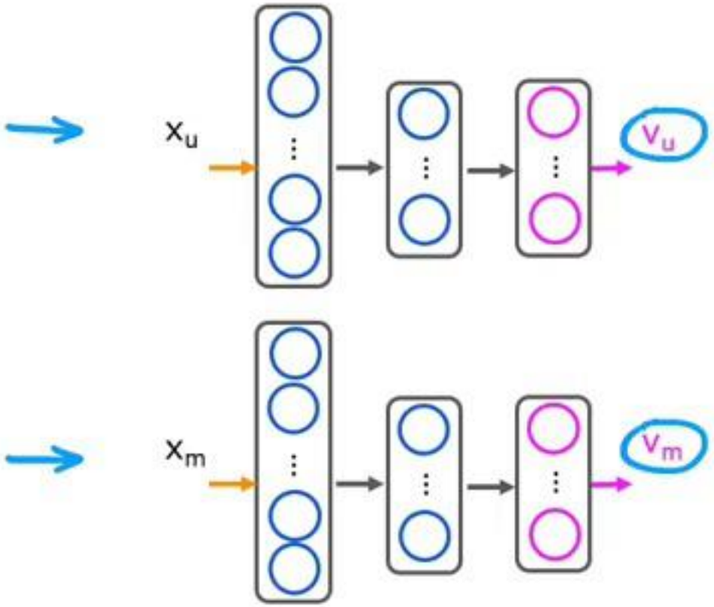
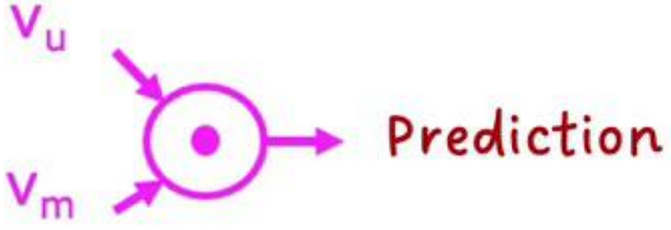
Content-based Filtering

TensorFlow Implementation




```
user_NN = tf.keras.models.Sequential([
    tf.keras.layers.Dense(256, activation='relu'),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(32)
])
```

```
item_NN = tf.keras.models.Sequential([
    tf.keras.layers.Dense(256, activation='relu'),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(32 )
])
```

```
# create the user input and point to the base network
input_user = tf.keras.layers.Input(shape=(num_user_features))
vu = user_NN(input_user)
vu = tf.linalg.l2_normalize(vu, axis=1)

# create the item input and point to the base network
input_item = tf.keras.layers.Input(shape=(num_item_features))
vm = item_NN(input_item)
vm = tf.linalg.l2_normalize(vm, axis=1)

# measure the similarity of the two vector outputs
output = tf.keras.layers.Dot(axes=1)([vu, vm])

# specify the inputs and output of the model
model = Model([input_user, input_item], output)

# Specify the cost function
cost_fn = tf.keras.losses.MeanSquaredError()
```